Learning by Accretion: Separating Trust, Knowledge, and Skill in Language Models — an Experimental Study at Toy Scale

Lam Shiu Ching

Independent researcher · August 31, 2026 · code: github.com/LamShiuChing/eureka

Abstract. Contemporary language models entangle three functions in a single set of weights: representing language, storing knowledge, and executing skills. We argue that several of their most persistent failures — confident hallucination, uncorrectable factual errors, catastrophic forgetting under continued training, and frozen knowledge cutoffs — are consequences of this entanglement rather than independent defects. We report a sequence of sixteen controlled experiments, conducted on a single consumer GPU using 10M–89M-parameter GPT models trained on synthetic fact worlds with fully known ground truth, that separate these functions into distinct mechanisms: (i) a *provenance channel* — per-token source embeddings — from which models learn source reliability without reliability labels, defeating a coordinated lying majority (97% vs. 29% control) and localizing trust per source-and-topic; (ii) an *external nearest-neighbor memory* that restores collapsed factual recall (4/20 → 20/20), supports instant knowledge editing, and provides zero-forgetting continual fact learning; (iii) *routed adapter modules* that add skills append-only, where naive continued training erases prior skills entirely; (iv) an autonomous controller that detects missing skills geometrically, synthesizes oracle-verified training data, validates or discards its own products, and consolidates verified module behavior into the base — in two cases exceeding the accuracy of the modules that taught it. We also report three instructive negative results: self-evaluation could not be probed, jointly trained, or induced with rare examples in the memorization regime, because a converged model's training set contains no failures to learn from. We synthesize these results into a design in which pretraining buys representational coverage, knowledge lives in curated stores, skills accrete as modules, and consolidation is periodic — and state the two limits no internal mechanism escaped: ground truth must enter from outside the corpus, and no adapter can build on representations the base lacks.

1 Introduction

A large language model is at once a grammar, an encyclopedia, and a toolbox, implemented in one undifferentiated parameter block. This paper examines the hypothesis that this entanglement — rather than insufficient scale — underlies four chronic failures: hallucination (facts reconstructed lossily from compressed weights), uncorrectability (no operation edits a single fact without side effects), catastrophic forgetting (new training overwrites old competence), and staleness (knowledge frozen at training time).

We adopt an unusual methodology: extremely small models (10M–89M parameters, character- or word-level) trained on synthetic fact worlds where ground truth, source reliability, and the model's exposure history are exactly known. This scale cannot demonstrate competitive capability; it is chosen because it renders every claim falsifiable in minutes and every failure attributable. Each experiment changes one variable against a control, and negative results are reported with diagnoses.

Contributions: (1) evidence that source trust is learnable from provenance metadata alone, without reliability labels, and strengthens markedly with scale; (2) a characterization of its failure surface — trust degrades linearly with anchor corruption and over-generalizes into unaudited topics; (3) a

three-part negative result locating why post-hoc self-evaluation fails in the memorization regime; (4) demonstrations of zero-forgetting factual learning via an external memory and append-only skill learning via routed adapters; (5) a closed autonomous loop — detect, synthesize, train, validate, register — and a consolidation ("sleep") procedure in which verified self-distillation transferred skills into the base at accuracies exceeding their teacher modules; (6) a working unified system with persistent conversational memory on an adopted open-weights core.

2 Experimental setup

All experiments use a fork of nanoGPT [8] (GPT-2-architecture decoder-only transformers [7]) extended with: an optional per-token *source-embedding* channel added to the input sum; optional zero-initialized bottleneck adapters after each block; an optional confidence head; and dataset-path flexibility. Unless stated, models are 6 layers, 384 dims (10.6M parameters), trained ~2,000 iterations (2–3 minutes on an RTX 4080 SUPER).

Fact worlds are generated programmatically: entities with attributes (colors, sizes; later real countries with capitals and continents), rendered as claims (`The color of X is Y.`) optionally attributed to sources via the provenance channel, and question–answer pairs (`Q: … A: …`) that serve as the supervised *anchor*. Held-out entities receive claims but no QA, so recall and trust for them must generalize. Evaluation is exact-match over greedy decoding; every table reports raw counts. Results are single runs without variance estimates; we flag this in §9.

3 Calibration, and the failure of post-hoc self-evaluation

Abstention is trainable through data incentives (E2). A baseline model trained only on known-entity QA fabricates an attribute for 100/100 entities it never saw — the toy form of hallucination, consistent with the incentive account of Kalai and Nachum [1]. Adding QA lines answering `unknown` for 400 *distractor* entities (disjoint from both known and held-out sets) taught a rule that generalized: fabrication on 100 never-seen entities fell to 30% with zero cost to known-fact recall (100/100). Detecting *conflicting* evidence generalized only weakly under the same approach (58% hits over a 32% false-alarm floor).

Self-evaluation is not learnable post hoc (E3 series). We attached a linear probe to the final hidden state, trained against on-the-fly labels ("was the model's argmax correct?"). Probed passively: AUC 0.458 between memorized and fabricated answers — chance; the model's own maximum softmax probability was equally blind (AUC 0.479, saturated at 1.000 for both classes). Letting the probe's gradients shape the trunk: AUC 0.521. Injecting corpus-rare "one-shot" QA lines intended to supply genuine failure states: null again — at ~64 effective epochs the model had memorized 37/50 of them, and the failure states evaporated. The diagnosis is structural: *a converged memorizer's training set contains no failures*, so correctness-supervised confidence has nothing to learn. Calibration signals must come from data the model cannot have absorbed — which is precisely what E2's distractors provided.

4 Provenance: learning whom to trust

Standard pretraining discards where text came from. We add per-token source embeddings (`wse`), so claims carry their origin, and ask whether reliability can be *learned* — never labeled — from coherence with a small QA anchor.

Coordinated lies (EP). 500 entities; per entity, two truthful sources assert the true color and three "liar" sources agree on one wrong color, so the majority is always the lie (9 vs. 6 lines after repetition). The anchor covers 400 entities. On 100 held-out entities the tagged model sided with the outvoted-but-reliable minority **97/100** (1 lie); an identical model without tags followed the majority (29 truth, 56 lie) and could not even fully fit the anchor (94/100), having no way to reconcile anonymous contradiction.

Compositional trust (EP2). With two sources each truthful on one topic and lying on the other, balanced votes, and both sources exactly 50% reliable globally, only trust conditioned on (source, topic) can succeed on both topics. Held-out accuracy was 85% and 84% — symmetric, ruling out both vote-counting and scalar source preference.

Failure surface (EP4a/EP4b). Corrupting a fraction of the anchor (its QA answers replaced by the coordinated lie) degraded held-out truth linearly — 100/92/69/46% at 0/10/25/50% poison — reaching a statistical coin flip exactly where the two source coalitions become equally anchor-consistent. Trust therefore requires majority anchor integrity but tolerates modest rot. Coverage is less forgiving: with the size topic given claims but no QA anywhere, size questions first failed *closed* (answered with colors — the answering format itself was never taught); after teaching the format with 20 source-neutral entities, the hole failed *open*: the model answered 80–82% of unaudited size questions with the lies of the source whose color record had earned its trust. Absent coverage, the model adopts the simplest anchor-consistent hypothesis — global trust — and credibility laundering follows.

Scale (EPX). At 89M parameters, word-level, over 5,012 entities with natural templates and plausible lies, a fully trained model recalled held-out facts — no QA supervision, balanced conflicting claims — at **191/200** (capitals) and **199/200** (continents), with zero and one lie adoptions respectively. The same design at 10M/character-level managed 4/20. Provenance-conditioned recall is an ability that *emerges with scale*; intermediate checkpoints show trust-consistent recall arriving before shared-label bindings converge.

experiment	condition	truth	lie	other
EP (heldout, n=100)	tagged	97	1	2
	untagged control	29	56	15
EP4a (n=100)	0 / 10 / 25 / 50% anchor poison	100 / 92 / 69 / 46	0 / 4 / 20 / 47	—
EP4b-v2 (size, n=100)	topic unaudited	5	80–82	13–15
EPX (heldout, n=200)	89M, word-level	191 / 199	0 / 1	9 / 0

Table 1. Provenance trust: main results. EPX cells: capitals / continents.

5 External memory: recall, editing, continual facts

EP3 (real facts, natural English, 10M/char) showed trust surviving naturalism (held-out lie rates 0–15%) while unsupervised *recall* collapsed — the model garbled exact strings it was never QA-taught. Following kNN-LM [2], we built a datastore of (hidden state → next token) pairs from trusted QA text and interpolated it into decoding, $p = \lambda p_{kNN} + (1-\lambda)p_{LM}$, with no retraining. Held-out recall rose from 4/20 to 20/20 (capitals) and 10/20 to 20/20 (continents) at $\lambda \geq 0.5$; training-set recall *improved* (27→30), the store correcting the model's own misses. Editing the datastore alone redirected answers

instantly (Jakarta→"Asuncion") with weights frozen. Notably λ=1.0 also succeeds: even with the LM distribution fully replaced, the network remains the query encoder — suggesting weights are better spent on understanding questions than on storing answers.

In a sequential-learning test (E5), naive continued training on 200 new facts destroyed recall of 200 old ones (100→15/100); mixing 25% replay preserved both (100/100); appending the new facts to the store matched replay (100/100 both) with *zero* gradient steps. For factual content, the store — not training — is the continual-learning mechanism.

6 Skills: adapters, interference, and routing

Skills, unlike facts, cannot be table entries. We test whether a new skill (two-digit addition, evaluated on held-out operand pairs so success requires the procedure, not answer memorization) can be added without erasing an existing one (fact QA), using zero-initialized bottleneck adapters (Houlsby et al. [3]; cf. LoRA [4]) trained with the base frozen.

Naive continued training erased the old skill entirely (100→0) while mastering the new (100/100). Frozen-base adapters first failed doubly, for a foundational reason: the base's corpus contained no digits, so digit embeddings were untrained noise the adapter could neither use nor repair — *the "capable base" requirement extends to token embeddings*. With benign digit exposure added to base pretraining, adapters learned genuine arithmetic through the frozen base (63/100 held-out) but still destroyed fact QA (100→0): an always-on adapter fires on every input, and a loss over only new-task data never penalizes interference. Freezing weights protects parameters, not behavior. Routing — engaging the adapter only on its task — restored the old skill completely (100/100) at unchanged new-skill accuracy. A second adapter (subtraction, 97/100) and a *learned* router — a 1,155-parameter linear probe on the frozen base's hidden state, trained on 450 task-labeled question prompts — routed 150/150 held-out questions, yielding a three-skill system (100/63/97) with nothing forgotten.

arm	old skill (facts)	new skill (heldout addition)
base	100	0
naive continued training	0	100
replay (25%)	100	100
adapter, always-on	0	63
adapter + routing	100	63

Table 2. Continual skill learning (E7), n=100 per cell.

7 Autonomy: self-teaching, consolidation, curiosity

Self-teaching (E8). A controller with no task knowledge monitors a query stream using distance-based novelty over base representations, with a self-calibrated threshold (maximum familiar-question distance ×1.1). In a 200-question stream containing 50 multiplications — a skill no module possessed — it flagged all 50 with 3 false alarms, inferred the pattern and operand ranges from the flagged queries, synthesized 6,000 examples answered by an executable arithmetic oracle, and trained an adapter. Its validation gate rejected the first attempt (36/100, below its 50 threshold), escalated capacity by policy (64→128 dims), accepted the second (73/100, old skills intact), and registered the class with the router. Final routed system: 100/63/100/74 across four skills.

Consolidation (E9). Bottleneck adapters do not merge arithmetically (nonlinear serial modules do not fold into weight matrices), so we consolidate behaviorally: each module answers thousands of fresh questions, the oracle keeps only verified answers (yields 60%, 99%, 75% — tracking each module's own accuracy), facts replay from the store, and a copy of the base continues training on the mixture. The bare consolidated base — adapters removed — scored 100/92/92/81 across the four skills. Two skills exceeded the modules that taught them (addition 63→92; multiplication 74→81): the oracle filtered the teachers' noise, so the full-capacity student learned from a cleaner curriculum than its teachers ever saw. Verified self-distillation is amplification, not merely transfer.

Curiosity (E10). Replacing the external query stream with self-directed exploration: the system surveys a question grammar, scores each region by failure rate on oracle-verifiable probes, and teaches itself the argmax. Unprompted, it found division at 0/30, trained an adapter to 100/100 validation, and on re-survey reported the frontier empty. Intrinsic motivation here is an objective function over verifiable failure, not an anthropomorphic property.

Unified system. The organs compose into a served system with persistent conversational memory (facts extracted from chat survive full process restarts via a human-readable JSON store) and were transplanted unchanged onto an adopted open-weights core (GPT-2 124M [7], loaded directly into the fork), where the tool layer visibly corrects the core's arithmetic hallucinations and the store answers personal facts the core cannot contain.

8 Synthesis: the accretion architecture

The results converge on a division of labor. *Pretraining buys representational coverage* — the floor beneath every later mechanism (E7's embedding lesson; the router's linear separability; the store's query encoding). *Knowledge belongs in curated, provenance-weighted stores* — recallable exactly, editable instantly, learnable continually (E5/E6), trusted per source and topic (EP–EPX). *Skills accrete as routed modules* (E7/E7b) and *consolidate periodically* into the core through verified self-distillation (E9), with growth of the core itself reserved for genuinely new primitives. Learning is append-only day to day; weights are rewritten only at consolidation, under replay protection, from verified curricula.

Two limits recurred in every guise and appear fundamental. First, the *anchor problem*: trust, calibration, and self-teaching are all learned relative to some verified signal — corrupt it and every downstream judgment inverts (EP4a); no internal mechanism manufactures ground truth, which must enter through execution, measurement, or curation. Second, the *representational floor*: no attachment can build on what the base cannot represent. Between these limits, remarkably much proved achievable with components of a few thousand to a few hundred thousand parameters.

9 Limitations

All results are single runs at toy scale on synthetic or template-rendered domains with exact-match evaluation; no variance estimates are reported, and several margins (e.g., E2b's 58 vs. 32) are within plausible run-to-run noise. Sources here are purely truthful or purely deceptive with static behavior; adversaries that earn trust before betraying it, Sybil provenance, and graded reliability are untested. The curiosity grammar and the escalation ladder are hand-designed. The memory was evaluated under favorable retrieval conditions (query format matching the store; paraphrase robustness and store poisoning untested). The autonomous loop closes only in oracle-verifiable domains; the general

verification hierarchy is future work. These experiments establish mechanisms and failure modes, not competitive capability.

10 Related work

The incentive account of hallucination follows Kalai and Nachum [1]; internal-state hallucination detection follows the semantic-entropy line of Farquhar et al. [5] and probe-based successors [6] — our E3 nulls delimit where such probes have nothing to read. The external memory is kNN-LM [2]; retrieval-during-pretraining (RETRO [9]) motivates trading parameters for stores. Adapters and LoRA [3,4] supply the module mechanism, here repurposed from efficiency to preservation, with routing recovering a mixture-of-experts structure from continual-learning constraints. Replay and catastrophic forgetting have half-century roots in the complementary-learning-systems account [10], which the store/consolidation cycle mirrors. Fully open training stacks (OLMo [11]) make the adopted-core path auditable end to end.

11 Conclusion

Treating trust, knowledge, and skill as separable organs around a frozen representational core converts several chronic LLM failures from mysteries into engineering surfaces with measured properties: trust is learnable and linearly anchor-bound; recall is a store property; forgetting is an architecture choice; self-teaching is safe exactly as far as verification reaches. The obvious next steps are scale (porting the organs onto a fully open modern core) and the verification hierarchy that would let autonomy extend beyond executable domains.

References

1. A. T. Kalai, O. Nachum. Why Language Models Hallucinate. arXiv:2509.04664, 2025.
2. U. Khandelwal, O. Levy, D. Jurafsky, L. Zettlemoyer, M. Lewis. Generalization through Memorization: Nearest Neighbor Language Models. ICLR, 2020.
3. N. Houlsby et al. Parameter-Efficient Transfer Learning for NLP. ICML, 2019.
4. E. J. Hu et al. LoRA: Low-Rank Adaptation of Large Language Models. arXiv:2106.09685, 2021.
5. S. Farquhar, J. Kossen, L. Kuhn, Y. Gal. Detecting hallucinations in large language models using semantic entropy. Nature 630, 2024.
6. J. Kossen et al. Semantic Entropy Probes: Robust and Cheap Hallucination Detection in LLMs. arXiv:2406.15927, 2024.
7. A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, I. Sutskever. Language Models are Unsupervised Multitask Learners. OpenAI, 2019.
8. A. Karpathy. nanoGPT. github.com/karpathy/nanoGPT, 2022.
9. S. Borgeaud et al. Improving Language Models by Retrieving from Trillions of Tokens. arXiv:2112.04426, 2022.
10. J. L. McClelland, B. L. McNaughton, R. C. O'Reilly. Why there are complementary learning systems in the hippocampus and neocortex. Psychological Review 102(3), 1995.
11. Ai2. Olmo 3. allenai.org/blog/olmo3, 2025.

Experiments designed and directed by the author and conducted with AI assistance (Claude, Anthropic) in a single session on 2026-08-31. All code, per-experiment write-ups, and exact reproduction commands: github.com/LamShiuChing/eureka.